AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

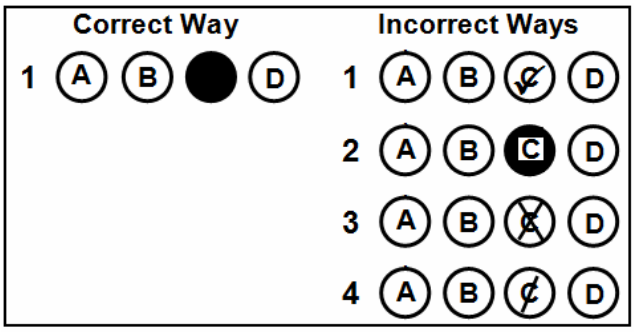
MAY 2014

Physics Paper I

Time: 35 minutes Marks: 25

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

1. The given diagram shows two persons holding the opposite ends of a rope.



If one person gives an up and down jerk to the rope, then which of the following types of wave is produced in the rope?

A. Periodic wave
B. Transverse wave
C. Longitudinal wave
D. Electromagnetic wave

2. If a pendulum makes 22 complete cycles of vibration in 11 seconds, then its frequency will be

A. 0.5 Hz
B. 1.0 Hz
C. 2.0 Hz
D. 4.0 Hz

3. The characteristic of sound which enables us to distinguish between normal sound and faint sound is its

A. pitch.
B. quality.
C. intensity.
D. frequency.

4. Which of the following is CORRECT about the speed of sound with respect to its medium?

	Speed of Sound (Slower to Fastest)		
A	Air	Steel	Water
B	Air	Water	Steel
C	Water	Air	Steel
D	Steel	Water	Air

5. Light bends towards normal when it enters from a rarer medium to a denser medium then its

A. speed decreases.
B. intensity increases.
C. frequency decreases.
D. wavelength increases.

6. The occurrence of total internal reflection will be possible when

I. angle of incidence is less than the critical angle.
II. angle of incidence is greater than the critical angle.
III. light enters from a denser medium to a rarer medium.

Which of the following is CORRECT about the given situation?

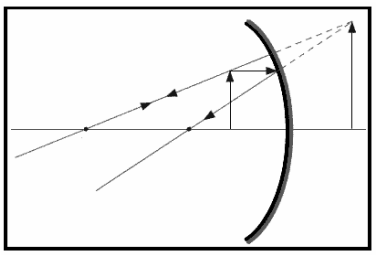
A. I only
B. II only
C. I and III
D. II and III

7. A long sighted person can see distant objects clearly but cannot see nearer objects distinctly.

In this defect, rays from nearby objects are focused

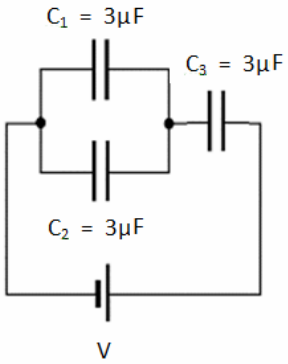
A. at the blind spot.
B. in front of retina.
C. behind the retina.
D. at the centre of retina.

8. Where should an object be placed to get an erect and virtual image using a concave mirror?



A. At *F*
B. Beyond *C*
C. Between *F* and *P*
D. Between *C* and *F*

9. The equivalent capacitance of the given circuit is



A. 11/3 μ F
B. 1/2 μ F
C. 2 μ F
D. 6 μ F

10. A hair comb becomes positively charged in use because

A. hairs have extra positive charge.
B. neutrons are created in the comb.
C. the comb loses electrons to the hair.
D. hairs gain extra protons from the comb.

11. Which of the following should be done to increase a circuit's resistance with an existing resistor?

	Component Added	Type of Connection
A	Resistor	Parallel
B	Resistor	Series
C	Capacitor	Series
D	Capacitor	parallel

12. Which of the following statements are CORRECT when there is no short circuit?

I Current flows through the earth wire.
II Current does not flow through the earth wire.
III All the currents from the neutral wire flow to the live wire.
IV All the currents from the live wire flow to the neutral wire.

A. I and III
B. I and IV
C. II and III
D. II and IV

13. Which of the following instruments is NOT common in AC generators and DC motors?

A. Slip rings
B. Armature
C. Magnetic poles
D. Carbon brushes

14. Which of the following devices is used to change the value of alternating voltage?

A. DC motor
B. Transformer
C. AC generator
D. Loud speaker

15. On which of the following principles does a transformer work?

A. Force induction
B. Strong induction
C. Mutual induction
D. Electrostatic induction

16. If a magnetic field changes in a conductor, then it produces

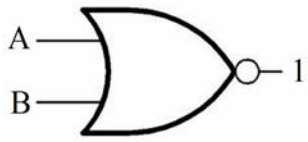
A. induced voltage.
B. capacitance.
C. inductance.
D. resistance.

17. Which of the following is used to carry electric charge across the space in a bulb?

A. Beta emission
B. Self induction
C. Mutual induction
D. Thermionic emission

18. Which of the following options CORRECTLY identifies the input value of the given NOR gate?

	Input A	Input B
A	0	1
B	1	0
C	1	1
D	0	0



19. Which of the following quantities maintains a constant rate of change in a particular span of time?

A. Digital quantities
B. Scalar quantities
C. Vector quantities
D. Analogue quantities

20. Which of the following electronic machines gives information after analyzing and arranging data in a very short interval of time?

A. Radio
B. Computer
C. Telephone
D. Fax machine

21. Which of the following represents 'X' in the given nuclear reaction?

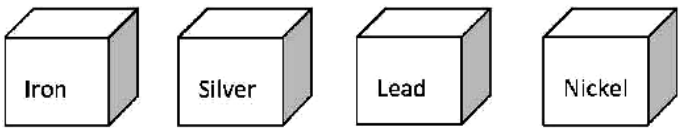
$${}_1H^1 + {}_1H^1 + {}_1H^2 \longrightarrow X + {}_{-1}e^0 + \text{energy}$$

A. Proton
B. Neutron
C. Beta particle
D. Alpha particle

22. Carbon dating of an ancient object determines

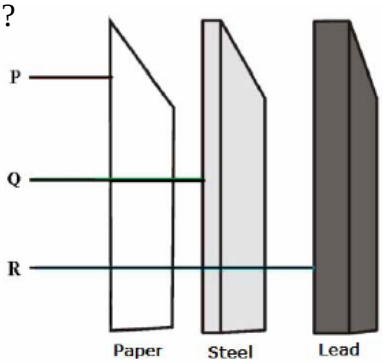
A. how long it lived.
B. its average life time.
C. the age when the object died.
D. how many years before it died.

23. Which of the following boxes is used to store radioactive material?



(A) (B) (C) (D)

24. In the figure below different sheets are placed in front of nuclear radiation. Which of the following ray/s can be identified as gamma rays?



A. P
B. Q
C. R
D. P and Q

25. The reaction which occurs on the surface of the sun is

A. fusion.
B. fission.
C. annihilation.
D. pair production.

END OF PAPER

Please use this page for rough work

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